mnirs v0.4.2.2

Muscle Near-Infrared Spectroscopy Processing and Analysis

Read, process, and analyse data from muscle near-infrared spectroscopy (mNIRS) devices. Import raw data from .csv or .xls(x) files and return a time-series data frame with metadata. Includes standardised methods for cleaning, filtering, and pre-processing mNIRS data for subsequent analysis, with a custom plot theme and colour palette.

Workflow

```
├──── read_mnirs()
│     ├──── create_mnirs_data()
│     └──── example_mnirs()
├──── resample_mnirs()
├──── replace_mnirs()
│     ├──── replace_invalid()
│     ├──── replace_outliers()
│     └──── replace_missing()
├──── filter_mnirs()
│     ├──── filter_ma()
│     └──── filter_butter()
├──── shift_mnirs()
├──── rescale_mnirs()
├──── plot.mnirs()
│     ├──── theme_mnirs()
│     ├──── palette_mnirs()
│     ├──── scale_colour_mnirs()
│     ├──── scale_fill_mnirs()
│     ├──── format_hmmss()
│     └──── breaks_timespan()
└──── extract_intervals()
```

Read

`read_mnirs()` — Import mnirs data from .csv or .xls(x)

Package-level functions

`example_mnirs()` — Paths to an included example file

`create_mnirs_data()` — Add *"mnirs"* class metadata to data

Resample

`resample_mnirs()` — Resample to a regular time grid

Clean

`replace_mnirs()` — Replace outliers, invalid, and missing data

Vector-level functions

`replace_outliers()` — Replace local outliers

`replace_invalid()` — Replace specified invalid values

`replace_missing()` — Interpolate over *NA* values

Filter

`filter_mnirs()` — Apply digital filter to mNIRS channels

Vector-level functions

`filter_butter()` — Apply a Butterworth digital filter

`filter_ma()` — Apply centred moving average smoothing

Transform

`shift_mnirs()` — Shift channels to a reference value

`rescale_mnirs()` — Rescale channels data range

Plot

`plot()` — Plot a data frame of class *"mnirs"*

`theme_mnirs()` — ggplot2 theme for mnirs plots

`palette_mnirs()` — Access the mnirs colour palette

`scale_colour_mnirs()` — Colour scale with mnirs palette

`scale_fill_mnirs()` — Fill scale with mnirs palette

`format_hmmss()` — Plot axis in time units

`breaks_timespan()` — Plot axis with time breaks

Extract

`extract_intervals()` — Detect events and extract intervals